Fabric Surface Defect Detection and Localization on AFID with Classical Texture Descriptors: from One-Class Texture References to a Supervised Multi-Scale Detector

Rinal Shah, Varun Hotani, Aadi Mehta, Pratik Agrawal
ECE501 Digital Image Processing, Group 15

Abstract—We detect and localize defects in woven fabric on the AITEX Fabric Image Database (AFID) using only classical image processing: illumination flattening, median filtering, CLAHE, GLCM, local binary patterns, Gabor filter banks, a per-pixel filter bank and the Hough transform. A first, one-class system modelled defect-free texture per fabric. Under a leak-free protocol it reached only 1.1 % pooled pixel IoU, and diagnosis showed why: within a strip its scores barely rank defects above background (AUROC 0.54), because AFID's defect-free and defective images differ globally. The final system keeps the classical descriptors but learns defect appearance from the dataset's masks, with gradient-boosted trees on 16×16 windows with strip-normalised, multi-scale context. It is evaluated by 5-fold cross-validation grouped by strip, with every threshold chosen on validation strips. It reaches 0.98 window AUROC, 62.3 % patch F1, 80.8 % image-level accuracy over 245 strips, and 20.9 % pooled pixel IoU (33.9 % when the single strip holding 48 % of all defect pixels is excluded).

Index Terms—fabric defect detection, texture analysis, GLCM, local binary patterns, Gabor filters, Mahalanobis distance, gradient boosting, conformal prediction, AFID

I. Introduction

Automatic inspection of woven fabric must find defects that are often a few pixels wide on a periodic background whose appearance changes with the weave, the yarn and the illumination. The AITEX Fabric Image Database (AFID) [1] provides 245 line-scan strips of 256×4096 pixels from seven fabrics: 140 defect-free strips and 106 defective strips with pixel masks for 12 defect classes. This project set out to detect and localize those defects with classical digital image processing only: hand-crafted texture descriptors and filter banks, with no learned convolutional features.

The work proceeded in two phases.

- Phase 1 (one-class, Weeks 3–7). Per-fabric texture references were fitted on defect-free strips only, and strips were scored by their deviation from them. This phase established the pipeline, the data handling and an honest evaluation protocol. It ended with a diagnosis: within a strip, its scores barely rank defects above background.
- Phase 2 (supervised, Week 8). The same descriptors, plus a per-pixel filter bank, train gradient-boosted trees on AFID's masks. Evaluation uses strip-grouped cross-validation, so no test strip ever contributes to training or to any threshold.

Contributions:

1) a reproducible, tested pipeline for AFID, including a complete account of the dataset's quirks
2) a quantified capture shift between AFID's defect-free and defective images, which bounds any defect-free-only detector
3) a classical multi-scale detector (window classifier at 16×16 with context, pixel filter bank, Hough line linking), evaluated leak-free with precision, recall, F1-score and IoU at pixel, window, patch and image level

II. Related Work

Gray-level co-occurrence matrices [2], local binary patterns [3] and Gabor filter banks [4], [5] are the standard classical texture descriptors, and all three have been applied to fabric inspection. Novelty detection against a model of normal texture, such as a Mahalanobis distance [6], a One-Class SVM [7] or an Isolation Forest [8], needs defect-free data only. Supervised alternatives need labelled defects; gradient-boosted trees [9], [10] are a strong classical choice for tabular descriptors. The Hough transform [11], [12] detects line structures. Conformal prediction [13] gives distribution-free guarantees on false-alarm rates.

III. Dataset and Evaluation Protocol

A. AFID

Strips are 256 rows (fabric travel) by 4096 columns (fabric width). Filenames encode nnnn_ddd_ff: image, official defect code, fabric code. Five quirks of the published archives are handled explicitly:

- a misnamed extra defect-free strip
- a defective strip of a non-existent fabric 08 without a mask
- two masks split into two files
- one empty mask

- one 256×3796 strip

The usable set is 105 defective strips with masks and 140 defect-free strips. Defect classes are highly imbalanced: 39 fuzzy balls, but a single knot, contamination and weft crack.

B. Protocol

All results use strip-level splits: overlapping windows of one strip never appear on both sides. The one-class phase holds out defect-free strips per fabric (12 fit, 4 calibrate, 4 image-level negatives). The supervised phase uses 5-fold cross-validation grouped by strip and stratified by (defective, fabric). Each outer training fold is split again, by strip, into fit (75 %) and validation (25 %) strips. Every threshold and configuration choice is made on validation strips by maximising F1, and the test strips are scored exactly once. Metrics are reported pooled (TP/FP/FN summed over strips) and as per-strip means. Undefined ratios are NaN, not zero.

IV. Method

A. Pre-processing

Each strip is cropped to the fabric region of interest: columns with mean below 245 and standard deviation above 2, minus a 40 px selvage margin. It is then illumination-flattened by subtracting a Gaussian blur with $\sigma = 50$ px, median-filtered (3×3) and contrast-normalised with CLAHE (clip 2.0, 8×8 tiles).

B. Texture Descriptors

For square windows at several sizes, the following blocks are computed:

- GLCM (30-D): five Haralick statistics at distances 1–3 over four angles (mean and spread across angles), on a min–max stretch quantized to 64 levels.
- LBP (28-D): uniform rotation-invariant histograms at $(P, R) = (8, 1), (16, 2)$.
- Gabor (24-D): the mean and standard deviation of magnitude responses of 12 filters ($\lambda = 4, 8, 16$ px; 0°–135°; one-octave bandwidth).

LBP codes and Gabor responses are computed once per strip and pooled per window. All blocks are cached per image and per window size.

C. Phase 1: One-Class Texture References

Per fabric, a regularised Gaussian model of defect-free 256×256 descriptors scores windows by Mahalanobis distance:

$$d_f(\mathbf{x}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu}_f)^\top (\boldsymbol{\Sigma}_f + \lambda \tfrac{\operatorname{tr} \boldsymbol{\Sigma}_f}{D} \mathbf{I})^{+} (\mathbf{x} - \boldsymbol{\mu}_f)}. \quad (1)$$

The threshold $\tau_f = \mu + k\sigma$ ($k = 2.5$) is set on held-out defect-free strips. Flagged windows are refined at 64×64 (optionally 32×32), and morphology produces the mask. An image-level decision uses a conformal p-value of $\max_i d_f / \tau_f$ against held-out defect-free strips. The ablations compared one-class SVM and Isolation Forest baselines, a pooled fixed reference against the per-fabric adaptive ones, strip-normalised scoring and an in-domain cross-validated reference.

TABLE I
Progression over the project (AFID; %, except AUROC). Patch F1 and IoU_μ are per-strip means; pixel F1 and IoU are pooled (TP/FP/FN summed over strips). Image-level AUROC uses 28 (Week 7) or all 140 (final) defect-free strips as negatives.

System	n	Patch F1	Pix. F1	IoU	IoU_μ	AUROC
Mid-sem (13 strips)	13	27.1	–	–	0.4	–
Week 5 one-class	105	14.0	0.7	0.3	0.3	0.675
Week 7 one-class	105	18.4	2.2	1.1	1.4	0.790
Supervised (final)	105	58.6	34.5	20.9	14.0	0.865

D. Phase 2: Supervised Multi-Scale Detector

Window stage. Every 16×16 window (stride 8) gets a 417-D row with four parts:

- its own GLCM, LBP and Gabor blocks
- the same blocks robust-z-scored within the strip (median/MAD), which cancels strip-wide offsets
- the strip-normalised blocks of the nearest 32×32, 64×64 and 256×256 windows (context), found by binary search on the window grid
- a fabric one-hot

A histogram gradient-boosting classifier [9], [10] (early stopping, at most 1,000 rounds) is trained on every defect window and on sampled defect-free windows, with the classes re-weighted. A second classifier on 32×32 windows provides an alternative map. Validation selects the 16×16 map, the 32×32 map or their mean.

Pixel stage. Inside candidate windows, each pixel is classified from 30 robust-normalised filter responses:

- intensity, and its deviation from 5/15/31 px medians
- local standard deviation over 5, 15 and 31 px
- weft- and warp-line responses: deviation from the column and row medians, smoothed along the yarn over 9 and 33 px
- Laplacian of Gaussian at $\sigma = 1, 2, 4$
- smoothed Sobel gradients
- the 12 Gabor magnitudes

Line linking. Probabilistic Hough segments [12] within 10° of the yarn directions are redrawn inside flagged windows only, to close gaps in thin detections.

Decisions. Validation chooses between the window-map mask and the pixel-stage mask, and sets the gate and pixel thresholds, the Hough settings, and the window, patch and image thresholds, each by maximising F1. Pooled IoU $= F_1/(2 - F_1)$ is maximised with it.

V. Results

Table I traces the project. The one-class system improved between Weeks 5 and 7 but stayed near 1 % pooled pixel IoU. The supervised detector, on the same

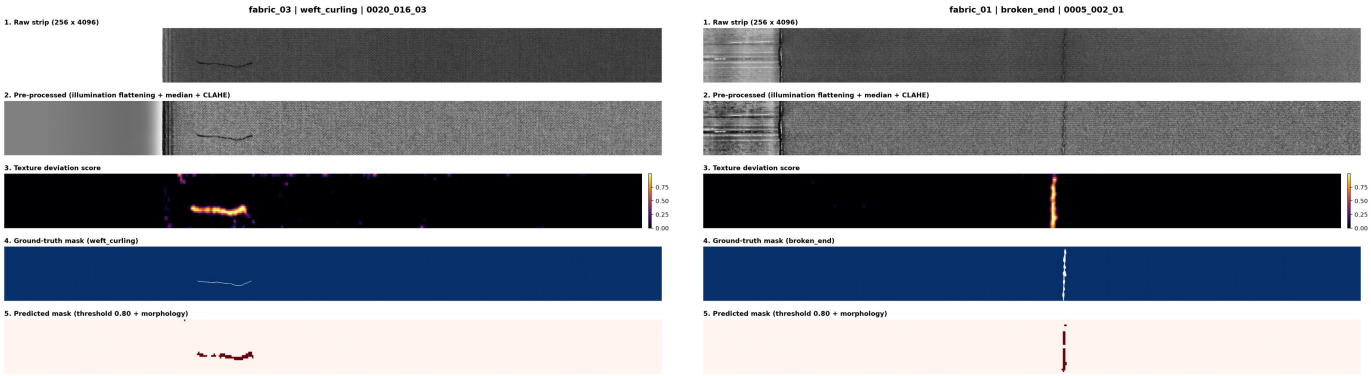


Fig. 1. Held-out predictions of the final detector. Left: weft curling (fabric 03). Right: broken end (fabric 01). Panels: raw strip, pre-processed strip, 16×16 texture-deviation map, ground truth, predicted mask.

TABLE II
Final supervised detector: precision, recall, F1 and IoU (%) at every level (pooled unless stated). †Excluding strip 0080_036_01, whose mask holds 48 % of all defect pixels. ±3 px: a prediction within 3 px of a defect pixel counts as correct.

Level	Prec.	Rec.	F1	IoU
Pixel, pooled	52.3	25.8	34.5	20.9
Pixel, pooled†	52.2	49.1	50.6	33.9
Pixel, ±3 px	65.3	29.2	40.4	–
Pixel, mean/strip	28.8	40.6	21.1	14.0
16^2 window	58.0	35.0	43.7	28.0
32^2 window	54.7	38.7	45.3	29.3
256^2 patch	60.3	64.3	62.3	45.2
Image (strip)	81.5	71.4	76.1	–

TABLE III
Ablation (%, except AUROC): supervised variants (each a full CV) and one-class systems, including the fixed global reference against the adaptive per-fabric one.

Variant	Pix. F1	IoU pooled	IoU mean	Img AUROC
final: 16x16 + 32x32 ensemble, pixel stage, Hough	34.5	20.9	14.0	0.865
32x32 windows only	33.4	20.1	15.1	0.872
16x16 + 32x32, no pixel stage	32.4	19.3	12.1	0.865
One-class, per-fabric	2.2	1.1	1.4	0.790
One-class, global	0.2	0.1	0.1	0.744
One-class SVM	2.2	1.1	1.3	0.697
Isolation Forest	1.8	0.9	1.0	0.669

descriptors, reached 20.9 %. Table II gives precision, recall, F1 and IoU at every level:

- 16×16 windows: AUROC 0.980, F1 43.7 %
- 256×256 patches: F1 62.3 %
- image level: accuracy 80.8 % (75 of 105 defective strips found, 17 false alarms in 140 defect-free strips), F1 76.1 %

On fabric 03, whose two capture sets match best, image accuracy is 88.2 % and image F1 90.3 %. Validation selected the 16×16 map in 4 of 5 folds (the 16/32 mean in one), the pixel stage in 2 and Hough linking in 3. In

TABLE IV
Final detector per official defect class (pooled pixel %, and image-level detection rate).

Defect	n	Prec.	Rec.	F1	IoU	Det.
broken_end	9	42.1	60.3	49.6	32.9	77.8
broken_pick	10	62.3	48.1	54.3	37.2	100.0
broken_yarn	8	34.3	65.6	45.0	29.1	100.0
contamination	1	72.8	72.8	72.8	57.2	100.0
crease	5	17.8	49.9	26.3	15.1	100.0
cut_selvage	9	22.7	23.0	22.9	12.9	44.4
fuzzy_ball	39	6.6	38.9	11.3	6.0	61.5
knot	1	–	0.0	0.0	0.0	0.0
nep	14	21.3	27.5	24.0	13.7	57.1
warp_ball	5	7.5	33.0	12.2	6.5	80.0
weft_crack	1	67.3	0.6	1.2	0.6	100.0
weft_curling	3	17.8	54.9	26.9	15.5	100.0

the ablation (Table III), removing the pixel stage lowers pooled pixel F1 from 34.5 to 32.4 %, and 32×32-only windows reach 33.4 %. Fig. 1 shows two held-out strips.

VI. Discussion

Adaptive vs. fixed. Among one-class systems, per-fabric (adaptive) references beat a single pooled (fixed) reference by a wide margin (Table III: 2.2 vs. 0.2 % pooled pixel F1). In the supervised detector, adaptivity takes the form of within-strip normalisation and a fabric code.

Capture shift. Normal windows of defective strips scored far above held-out defect-free strips for five fabrics, so no defect-free-only model could set a threshold that worked on both [14]. Strip-normalised features and defect labels remove that obstacle.

Limitations.

- Pooled pixel metrics are dominated by one strip. The weft crack 0080_036_01 holds 48 % of all defect pixels. It is detected at image level, but only 0.6 % of its pixels are recovered.
- Thin defects are delineated a few pixels off. F1 rises from 34.5 to 40.4 % within ±3 px.
- Small blobs are found but not outlined. Fuzzy balls (39 strips) have window AUROC 0.97 but 6 % pixel IoU.

- Several classes have a single example. Fabric 05 has one defective strip, so per-class results there are anecdotal.
- Supervision requires pixel masks for training.

VII. Conclusion

Classical texture descriptors are sufficient to find fabric defects on AFID, provided the decision is learned from labelled defects rather than from defect-free data alone. Evaluated without leakage, a gradient-boosted detector on 16×16 GLCM, LBP and Gabor windows with strip-normalised multi-scale context reaches 0.98 window AUROC, 62 % patch F1 and 81 % image accuracy. Exact pixel delineation of thin and small defects (21–34 % pooled IoU) is the main remaining limitation.

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